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Emerging Knowledge Trend in Statistical Research: A Content-Based Analysis Using Covariate-Assisted Dynamic Topic Model

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ABSTRACT

Exploring the emerging knowledge trends within a particular discipline is of great interest to scientific researchers. It helps researchers to understand the historical development of their disciplines and to guide their future research directions. In this work, we focus on the fast-developing discipline, statistics, to investigate its knowledge trend and emerging topics in statistical research. To this end, we collect publications in top-tier statistical journals and statistical-related conferences, and develop a covariate-assisted dynamic topic model (CDTM). It captures the dynamic evolution of topics in statistical publications and also finds the time-varying effects of covariates on topic discussions. To estimate CDTM, a variational inference procedure is applied. The theoretical properties are studied and finite-sample performance is evaluated through simulation experiments. Last, we apply CDTM to the collected academic data. We highlight the advantages of CDTM over other alternative methods and uncover topics that characterize the evolution of statistical knowledge over the past four decades. We also observe that certain research topics in statistics are increasingly aligning with advancements in artificial intelligence. Based on these findings, we gain valuable insights into the historical progression of statistical research, which enables us to better anticipate future trends and guide innovation in the field. Supplementary materials for this article are available online, including a standardized description of the materials available for reproducing the work.

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1. Introduction

The research interests of disciplines are always evolving, driven by technological advancements, societal needs, and emerging challenges. Understanding these shifts is crucial, as it allows researchers to stay informed about key developments within their fields. In this work, we focus on one of the fastest-developing disciplines, statistics, which emerged during the late 19th and early 20th centuries (Walker 1929). In the past century, many statistical branches were founded, including high-dimensional data analysis, time series analysis, Bayesian analysis, regression methods, variable selection, experimental design, and semi- and nonparametric theory, etc. Statistics has also been deeply integrated with other disciplines, such as biological and medical sciences, economics, and finance. In recent years, statistics has entered the artificial intelligence (AI) era, with an increasing number of subfields closely related to modern data science. Therefore, it is of great interest to explore the developing trends and emerging topics of statistics.

To explore the emerging knowledge trend of a particular discipline, one option is to analyze the publications of top-tier journals in that discipline. It is believed that the published papers are reflections of the most popular and active research topics, which are connected through different forms of citations. Therefore, the publications in top-tier journals can be used to explore the emerging knowledge trend of a particular discipline. In this regard, Ma and Yu (2010) detected several important research topics and concepts through an intellectual map which was built upon the citation networks of publications in top-tier journals. Liu, Jiang, and Ma (2013) developed a collective dynamic model for biological knowledge networks, aiming to identify the evolution mechanism of research topics over time. In the field of statistics, Ji et al. (2022) employed coauthor and co-citation networks to construct a research map of statisticians. Zhang et al. (2024) used co-occurrence networks to uncover popular keywords and research interests in the field of statistics.

Note that the published papers are mainly unstructured text documents. Therefore, techniques specifically designed for text mining can be used. Toward this goal, one suite of classical methods is topic models, which aim to find latent topics underlying a document corpus. The basic topic model is latent Dirichlet allocation (LDA, Blei, Ng, and Jordan 2003). It assumes that each document follows a distribution over a set of topics and each topic follows a distribution over the vocabulary. With these assumptions, a three-level probabilistic model can be established and estimated. The key advantage of topic models lies in their ability to represent a document with a low-dimensional and interpretable vector, that is, the document's topic distribution. Recently, with the development

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of AI, neural topic models (NTMs, Devlin et al. 2019; Dieng, Ruiz, and Blei 2019; Wu et al. 2024) and large language models (LLMs, Pham et al. 2024) have become popular in text mining. While NTMs and LLMs-based approaches perform remarkably well in many downstream tasks such as text classification and information retrieval, their training typically requires substantial amounts of data and computational resources. Prompting seems an easy way to adapt LLMs to specific tasks without fine-tuning or training. However, the results often lack consistency and transparency. In addition, these methods generally do not offer well-quantified interpretability for textual content. In contrast, probabilistic topic models can provide interpretable and quantifiable representations of texts, enabling structured analysis and inference. Lin et al. (2025) also found that ChatGPT, one representative of LLMs, eventually reverted to using traditional statistical methods for topic modeling. Therefore, in this work, we rely on probabilistic topic models to explore the emerging knowledge trends in statistics, while also leveraging LLMs to assist in the analysis. A more detailed discussion can be found in Appendix S3.3 of the supplementary materials.

To investigate the emerging knowledge trends from dynamic publications, one natural method is the dynamic topic model (DTM, Blei and Lafferty 2006). The main idea is to use a Gaussian process to capture the evolution of research topics. Other extensions of DTM include continuous-time DTM (Wang, Blei, and Heckerman 2008), embedded DTM (Dieng, Ruiz, and Blei 2019), Rolling LDA (Rieger et al. 2022), joint DTM for two dynamic text corpora (Zhu et al. 2022), and TOPIC-PYP for textual change-point detection (Wang et al. 2025). However, most existing dynamic topic models do not incorporate covariate information, which has been widely considered in static models. One typical work is the structural topic model (STM, Roberts, Stewart, and Airoldi 2016), which embeds external features into the generative process of documents. By incorporating covariate information, STM allows for more interpretable topics, as it links document-level metadata (such as time, location, or sentiment) to the topic distributions. However, STM is a static model, meaning it does not account for temporal or dynamic changes in the topic distributions over time.

To investigate emerging knowledge trends from published articles, our preliminary analysis (in Section 2) indicates that covariates do play a crucial role. This is because the topics of publications are often influenced by various factors, such as authors and publication venues. For instance, compared to theoretical topics, those related to real-world applications tend to involve a larger number of authors. Additionally, different publication venues may have their own preferences or styles, which can result in varying topic inclinations. Therefore, in this work, we incorporate covariate information to explore the knowledge trends in statistics and develop a covariate-assisted dynamic topic model (CDTM). Specifically, we use a Gaussian process to model the dynamic distribution of topics over words across time and apply a logistic-regression framework to incorporate the effects of covariates on the distribution of documents over topics. The covariate effects are also allowed to change over time. We estimate CDTM via variational inference, theoretically show that the estimates are consistent under regularity conditions, and demonstrate its empirical performance with synthetic documents. Finally, we apply CDTM to publications in top-tier statistical journals and statistics-related conferences to explore the knowledge trends in statistics over the past decades.

This article is organized as follows. In Section 2, we introduce the dataset and conduct some preliminary analysis, which motivate our methodology development. In Section 3, we introduce CDTM and discuss its estimation method and theoretical properties. In Section 4, we apply our method to statistical publications, exploring the emerging knowledge trend of statistics. We conclude this article with brief discussions in Section 5.

2. Data and Motivation

2.1. Data Collection and Pre-Processing

To reflect the emerging knowledge trend, we focus on top-tier statistical publications, which consist of two parts. The first part includes four internationally recognized, top-ranking statistics journals. They are, respectively, *Journal of the American Statistical Association* (JASA), *Annals of Statistics* (AOS), *Journal of the Royal Statistical Society Series B: Statistical Methodology* (JRSSB), and *Biometrika* (BKA). With the rapid advancement of artificial intelligence, many statisticians are engaging in AI-related research and publishing their findings in leading computer science conferences. Therefore, to ensure the completeness of our analysis, we incorporate a second data source, that is, the computer science conferences where statisticians actively contribute and disseminate their work. To this end, we consider three computer science conferences. They are, respectively, *Conference on Neural Information Processing Systems* (NeurIPS), *International Conference on Machine Learning* (ICML), and *International Conference on Artificial Intelligence and Statistics* (AISTATS).

We collect papers published by the four statistic journals from 1980 to 2024. However, for the three computer science conferences, we gather papers only from 2014 to 2024 due to data availability constraints, as earlier records are not unavailable. For each academic paper (in both journals and conferences), we collect its authors, title, publication venue, publication year, and abstract. The total numbers of papers in each publication venue are: 7590 in JASA, 5339 in AOS, 2352 in JRSSB, 3971 in BKA, 14044 in NeurIPS, 10667 in ICML, and 3567 in AISTATS. Note that the computer science conferences contain not only statistics-related articles but also many that are unrelated to statistics. Therefore, we implement a two-step pre-selection procedure to filter statistics-related articles from the three conferences. The first step is a preliminary filtering step to select articles potentially relevant to statistics. Given the uncertainty of the first filtering step, a subsequent validation step is conducted to ensure accuracy. The detailed procedure is shown in Appendix S5.1 of the supplementary materials. After this pre-selection process, the number of papers selected from each conference is as follows: 4285 (30.5% of 14,044) in NeurIPS, 2750 (25.8% of 10,667) in ICML, and 1448 (40.6% of 3567) in AISTATS.

2.2. Descriptive Analysis of Publication Trends and Impact

We present some descriptive analysis of publication trends and impact. We first focus on the publication volumes of the four statistical journals and three computer science conferences. For the conferences, early publication counts (prior to 2014) are

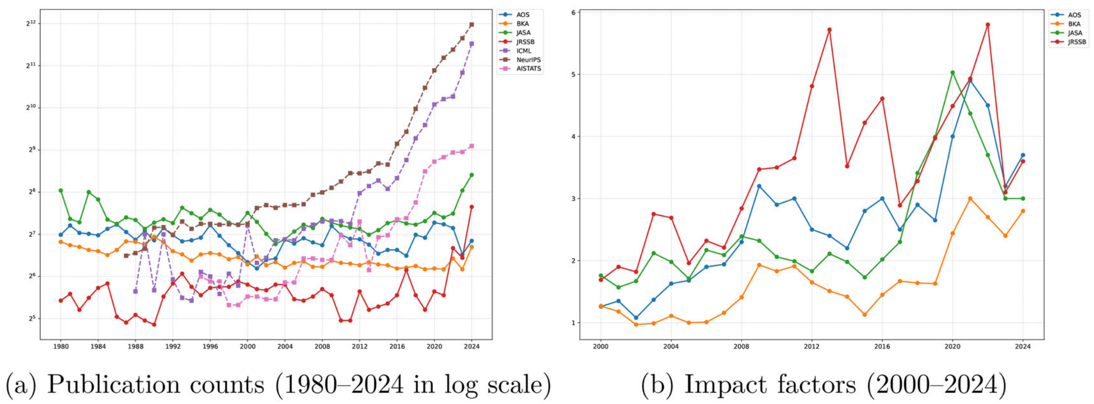


Figure 1. The publication counts and impact factors (the y-axis) over the years.

Table 1. Top-cited papers (Google Scholar, as of Oct 25, 2025) from four statistical journals published from 1979 to 2015, listed in descending order by total citation counts.

	Title	Year	Journal	Citation(total)	Citation (per year)
1	Controlling the false discovery rate: a practical and powerful approach to multiple testing	1995	JRSSB	122793	4093
2	Regression shrinkage and selection via the lasso	1996	JRSSB	66261	2285
3	Distribution of the estimators for autoregressive time series with a unit root	1979	JASA	43328	942
4	The central role of the propensity score in observational studies for causal effects	1983	BKA	43155	1028
5	Greedy function approximation: A gradient boosting machine	2001	AOS	37319	1555
6	Testing for a unit root in time series regression	1988	BKA	29251	791
7	Bootstrap methods: Another look at the Jackknife	1979	AOS	27974	608
8	Regularization and variable selection via the elastic net	2005	JRSSB	24839	1242
9	Longitudinal data analysis using generalized linear models	1986	BKA	22672	581
10	Bayes factors	1995	JASA	21452	715
11	Bayesian measures of model complexity and fit	2002	JRSSB	15904	691
12	Robust locally weighted regression and smoothing scatterplots	1979	JASA	15288	332
13	A proportional hazards model for the subdistribution of a competing risk	1999	JASA	15136	582
14	Ideal spatial adaptation by wavelet shrinkage	1994	BKA	13973	451
15	Least angle regression	2004	AOS	13217	629
16	The control of the false discovery rate in multiple testing under dependency	2001	AOS	12641	527
17	Multivariate adaptive regression splines	1991	AOS	12468	367
18	Variable selection via nonconcave penalized likelihood and its oracle properties	2001	JASA	11531	480
19	A test of missing completely at random for multivariate data with missing values	1988	JASA	11455	310
20	Sampling-based approaches to calculating marginal densities	1990	JASA	10451	299

collected from publicly available reports. As shown by Figure 1(a), the annual output of the three computer science conferences has grown exponentially since 2010, whereas the number of publications in the four statistical journals has remained relatively stable over the same period. Another key difference worth noting lies in the editorial policies of computer science conferences versus statistical journals. The three conferences typically maintain acceptance rates around 20%–30% (Li 2025). In contrast, statistical journals often aim to maintain a consistent number of publications annually, which makes them become more selective as submission volumes rise. For instance, JRSSB received about 200 submissions in 1990, with an acceptance rate of roughly 25% (Kent and Smith 1991). By 2020, submissions had increased to approximately 600, while the acceptance rate had dropped to around 8% (Delaigle and Wood 2021). This trend is not unique to JRSSB; JASA also reported acceptance rates of about 10% in recent years (American Statistical Association 2022). In contrast to the higher acceptance rates of conferences, this low acceptance rate in statistical journals has led to a growing shift of top research contributions toward computer science conferences.

Next, we focus on the impact of statistical publications. Figure 1b presents the impact factors of the four statistical journals, as reported by Clarivate. Note that impact factors for conferences are not provided by Clarivate and are thus omitted. Figure 1b reveals that the impact factors have some fluctuations over time, but the overall trend shows a general upward trajectory, indicating a gradual increase in the influence and recognition of these journals within the academic community.

To further investigate the impact of statistical publications, we analyze the highly cited papers. Pearson et al. (2025) analyzed the most cited publications of the 21st century and found that some of them are closely related to statistics, such as random forests, meta-analyses, and statistical software. Motivated by their findings, we conduct a similar analysis of highly cited papers in the field of statistics. To facilitate efficient access to citation data, we collect article-level citation counts in bulk from Clarivate. However, since Clarivate only indexes journal articles and not conference proceedings, our analysis is limited to journal publications. We then select the top 100 most-cited articles based on Clarivate data and manually retrieve the corresponding citation counts from Google Scholar, as it offers broader coverage and more up-to-date citation statistics. The final ranking and analysis are based on the citation counts obtained from Google Scholar. In addition, to provide a comparison between earlier and more recent highly cited papers, we divide our analysis into two time spans: pre-2015 and post-2015.

Table 1 shows the papers exceeding 10,000 citations by Google Scholar published pre-2015. As shown in Table 1, the most cited article is Benjamini and Hochberg (1995), which has

Table 2. Top cited papers (Google Scholar, as of Oct 25, 2025) from four statistical journals published after 2015, listed in descending order by citation counts per year.

	Title	Year	Journal	Citation(total)	Citation (per year)
1	Estimation and inference of heterogeneous treatment effects using random forests	2018	JASA	4183	598
2	Cross-validation: What does it estimate and how well does it do it?	2024	JASA	526	526
3	Generalized random forests	2019	AOS	2786	464
4	Surprises in high-dimensional ridgeless least squares interpolation	2022	AOS	1120	373
5	Visualizing the effects of predictor variables in black box supervised learning models	2020	JRSSB	1672	334
6	Quasi-oracle estimation of heterogeneous treatment effects	2021	BKA	1125	281
7	Nonparametric regression using deep neural networks with ReLU activation function	2020	AOS	1262	252
8	Gaussian differential privacy	2022	JRSSB	723	241
9	Making sense of sensitivity: extending omitted variable bias	2020	JRSSB	1152	230
10	Simple local polynomial density estimators	2020	JASA	1112	222

a total of 117,386 citations, with an average of 3913 citations per year. The pre-2015 era also featured foundational methodological innovations, such as Markov chain Monte Carlo (MCMC, Gelfand and Smith 1990), causal inference (Rosenbaum and Rubin 1983), Lasso (Tibshirani 1996), and bootstrap (Efron 1979). These works primarily addressed theoretical and computational challenges in data analysis, establishing frameworks that transcended disciplinary boundaries and had widespread applications in economics, biology, and medicine.

We then analyze the most cited papers (excluding review papers) published in these four journals since 2015, normalizing citation counts by publication year to account for temporal influence. As shown by Table 2, recent highly cited papers in statistical journals reflect the field's growing alignment with the challenges and opportunities brought by big data and artificial intelligence. Many of these works focus on developing scalable and interpretable methods for complex models. For example, Apley and Zhu (2020) proposed intuitive tools for explaining black-box models, while Cinelli and Hazlett (2020) offered practical sensitivity analysis frameworks. Advances in classical techniques such as cross-validation (Bates, Hastie, and Tibshirani 2024) and density estimation (Cattaneo, Jansson, and Ma 2020) have also contributed to making statistical methods more robust and widely applicable. In parallel, increasing attention has been paid to issues like privacy preservation (Dong, Roth, and Su 2022) and distributed learning (Jordan, Lee, and Yang 2019), reflecting the field's adaptation to modern computational and ethical demands. These trends are in line with recent reflections by leading scholars in statistics and data science. As highlighted in a panel discussion by Lin et al. (2025), there is growing consensus that statistics must not only respond to the rise of AI but also help shape its development. It offers rigor, interpretability, and uncertainty quantification, especially in areas where black-box models often fall short.

2.3. Descriptive Analysis for Textual Information

To gain an initial understanding of research topics in statistics, we focus on the textual information of abstracts. Some commonly used preprocessing steps are performed on the abstracts. The non-letter symbols are removed and all letters are converted to small cases. Then, we remove the stop words, as defined by the *Natural Language Toolkit* (NLTK, Bird and Loper 2004). To make the textual information more interpretable, we segment words for all abstracts using the *word dictionary model* (WDM, Deng et al. 2016). WDM treats each sentence as a concatenation of latent phrases drawn from a probabilistic dictionary. Since

Table 3. χ^2-test p-values for term occurrences across different covariates.

Words	*PublicationVenues*	*NumberofAuthors*
algorithm	0.000	0.000
application	0.004	0.040
Bayesian	0.020	0.480
dimension reduction	0.000	0.047
high dimensional	0.000	0.000
hypothesis test	0.000	0.054
LASSO	0.000	0.067
multiple test	0.000	0.090
neural network	0.000	0.000
transformer	0.000	0.003

phrase boundaries are not explicitly marked in English, WDM treats the task as a missing data problem and applies an expectation maximization (EM) algorithm to infer these boundaries. The resulting vocabulary thus includes both single words and multi-word phrases. After preprocessing, the total number of terms (including both words and phrases) is $V = 10004$. For simplicity, we still refer to all terms in the vocabulary as "words" in the subsequent analysis to align with common usage.

We conduct a detailed analysis of the textual information contained in the article abstracts. Specifically, we examine the top 10 most frequent words across different publication venues, analyze how term frequencies vary with the number of authors, and track the temporal evolution of representative words over time. The detailed results are present in Appendix S5.2 of the supplementary materials. Furthermore, we select 10 frequently used words as examples to statistically test their association with specific covariates (i.e., the publication venues and the number of authors). As shown in Table 3, at the significance level of $\alpha = 0.05$, all words show significant relations with publication venues, and some words are significant associated with the number of authors. These results imply that different covariates exhibit distinct preferences for words.

Based on the above analyses, we observe that the thematic focus of statistical publications has been continuously evolving. This evolution is, in part, associated with covariates such as publication venues and patterns of author collaboration. Notably, the direction of change increasingly aligns with trends in artificial intelligence and data-driven applications. Motivated by these observations, we aim to answer the following questions:

(Q1) How have key topics and methodological advancements in statistical research evolved over the past four decades?
(Q2) How has the field of statistics evolved in response to the rise of AI, and how do historical research developments connect with modern data science?

(Q3) What role do covariates (such as publication venues, author collaboration patterns) play in shaping statistical research trends?

These questions drive the methodological development outlined in Section 3. By addressing these questions, we aim to offer valuable insights for researchers, educators, and policymakers, enabling them to better understand past developments, anticipate future trends, and foster innovation in statistical methodology.

3. Methodology Development

3.1. Model Setup

Suppose we have a collection of documents $\mathcal{C} = \{\mathcal{C}_1, \mathcal{C}_2, \ldots, \mathcal{C}_T\}$, where $\mathcal{C}_t$ is collected at time t and contains D_t documents for $t = 1, 2, \ldots, T$. Let $D = D_1 + \cdots + D_T$, which stands for the total number of documents. The dth document in $\mathcal{C}_t$ contains $N_{d,t}$ words, which are represented by $\mathbf{w}_{d,t} = \{w_{d,t,1}, \ldots, w_{d,t,N_{d,t}}\}$. To characterize the semantic meanings of $\mathcal{C}$, we assume that there are a total number of K topics underlying the document corpus. The kth topic at time t is characterized with $\boldsymbol{\beta}_{k,t} = (\beta_{k,t,1}, \ldots, \beta_{k,t,V})^\top \in \mathbb{R}^V$, a probability vector over a vocabulary of V distinctive words. By definition, $\beta_{k,t,1} + \cdots + \beta_{k,t,V} = 1$. We allow the probability vector $\boldsymbol{\beta}_{k,t}$ to depend on t, which indicates that the topics underlying the document corpus may change over time. In other words, $\boldsymbol{\beta}_{k,t}$ characterizes the dynamic pattern of the kth topic over time t.

Next we construct a covariate-assisted dynamic topic model (CDTM). Following Blei and Lafferty (2006), we first use the Gaussian process to model the dynamic changes of the topic-word distributions $\boldsymbol{\beta}_{k,t}$s. To be precise, let $\boldsymbol{\tau}_{k,t} = (\tau_{k,t,1}, \ldots, \tau_{k,t,V})^\top \in \mathbb{R}^V$, which is a multinomial logit transformation of the topic vector $\boldsymbol{\beta}_{k,t}$. Define $\mathrm{softmax}(\boldsymbol{\tau}_{k,t}) = \exp(\boldsymbol{\tau}_{k,t}) / \sum_{v=1}^{V} \exp(\tau_{k,t,v})$, where $\mathrm{softmax}(\cdot)$ is further referred to as the softmax function. We assume that

$$\boldsymbol{\beta}_{k,t} = \mathrm{softmax}(\boldsymbol{\tau}_{k,t}), \text{ where } (\boldsymbol{\tau}_{k,t} \mid \boldsymbol{\tau}_{k,t-1}) \sim \mathrm{Normal}(\boldsymbol{\tau}_{k,t-1}, \sigma_\tau^2 \boldsymbol{I}_V), \tag{1}$$

where σ_τ^2 stands for the variance parameter and $\boldsymbol{I}_V$ is the $V \times V$ identity matrix. According to (1), we can capture the temporal evolution of the topics, where later time points are related to their preceding counterparts.

Next we focus on the document-topic probability distributions, which allow for inclusion of some well-defined covariates. Assume the dth document at time t is associated with the kth topic with probability $\theta_{d,t,k}$, for $k = 1, \ldots, K$. The document-topic probability vector is thus defined by $\boldsymbol{\theta}_{d,t} = (\theta_{d,t,1}, \ldots, \theta_{d,t,K})^\top \in \mathbb{R}^K$. By definition, $\theta_{d,t,1} + \cdots + \theta_{d,t,K} = 1$. Let $\boldsymbol{\eta}_{d,t}$ denote the vector of $\boldsymbol{\theta}_{d,t}$ after multinomial logit transformation. Note that we allow $\boldsymbol{\theta}_{d,t}$ to depend on some document-level covariates (e.g., the publication venues). Specifically, let $\boldsymbol{x}_{d,t}$ be a p-vector of covariates associated with the dth document at time t. We assume

$$\boldsymbol{\theta}_{d,t} = \mathrm{softmax}(\boldsymbol{\eta}_{d,t}), \text{where } (\boldsymbol{\eta}_{d,t} \mid \boldsymbol{\Gamma}_t) \sim \mathrm{Normal}(\boldsymbol{\Gamma}_t^\top \boldsymbol{x}_{d,t}, \boldsymbol{\Sigma}_t) \text{ and } \boldsymbol{\Gamma}_t = (\boldsymbol{\gamma}_{1,t}, \ldots, \boldsymbol{\gamma}_{K,t})^\top. \tag{2}$$

The random vector $\boldsymbol{\gamma}_{k,t}$ stands for the covariate effect at time t. We further assume all $\boldsymbol{\gamma}_{k,t}$s are independent with $\boldsymbol{\gamma}_{k,t} \sim \mathrm{Normal}(\tilde{\boldsymbol{\gamma}}_k, \sigma_k^2 \boldsymbol{I}_p)$. Here σ_k^2s and $\boldsymbol{\Sigma}_t$s are hyperparameters, $\boldsymbol{I}_p$ is the $p \times p$ identity matrix. For a noninformative prior on $\tilde{\boldsymbol{\gamma}}_k$, we take $p(\tilde{\boldsymbol{\gamma}}_k) \propto 1$.

We assume the words in each document follow a two-step generation process. Suppose the topics are labeled as $\{1, 2, \ldots, K\}$. Recall $\mathbf{w}_{d,t,n}$ is the nth word in the dth document, which is collected at time t. In the first step, a topic indicator $\mathbf{z}_{d,t,n}$ is generated from multinomial distribution with parameter $\boldsymbol{\theta}_{d,t}$, denoted with $\mathrm{Multinomial}(\boldsymbol{\theta}_{d,t})$. In the second step, the word $\mathbf{w}_{d,t,n}$ is generated from $\mathrm{Multinomial}(\boldsymbol{\beta}_{k,t})$, given $\mathbf{z}_{d,t,n} = k$. In symbols, we have $\mathbf{z}_{d,t,n} \sim \mathrm{Multinomial}(\boldsymbol{\theta}_{d,t})$; and $(\mathbf{w}_{d,t,n} \mid \mathbf{z}_{d,t,n} = k) \sim \mathrm{Multinomial}(\boldsymbol{\beta}_{k,t})$. CDTM consists of (1) and (2), whose detailed data generation process is present as follows.

1. For time $t = 1, 2, \ldots, T$:
    - (a) For $k = 1, \ldots, K$, draw realizations from the topic distribution $(\boldsymbol{\tau}_{k,t} \mid \boldsymbol{\tau}_{k,t-1}) \sim \mathrm{Normal}(\boldsymbol{\tau}_{k,t-1}, \sigma_\tau^2 \boldsymbol{I}_V)$ to obtain $\boldsymbol{\beta}_{k,t} = \mathrm{softmax}(\boldsymbol{\tau}_{k,t})$.
2. For each document $d = 1, 2, \ldots, D$:
    - (a) Draw realizations from the document distribution $(\boldsymbol{\eta}_{d,t} \mid \boldsymbol{\Gamma}_t) \sim \mathrm{Normal}(\boldsymbol{\Gamma}_t^\top \boldsymbol{x}_{d,t}, \boldsymbol{\Sigma}_t)$, where $\boldsymbol{\Gamma}_t = (\boldsymbol{\gamma}_{1,t}, \ldots, \boldsymbol{\gamma}_{K,t})^\top$, to obtain $\boldsymbol{\theta}_{d,t} = \mathrm{softmax}(\boldsymbol{\eta}_{d,t})$.
    - (b) For each word $n = 1, 2, \ldots, N_{d,t}$:
        - i. Draw a topic assignment from a multinomial distribution $\mathbf{z}_{d,t,n} \sim \mathrm{Multinomial}(\boldsymbol{\theta}_{d,t})$. Suppose the realization is k.
        - ii. Given $\mathbf{z}_{d,t,n} = k$, draw word $\mathbf{w}_{d,t,n}$ from multinomial distribution $(\mathbf{w}_{d,t,n} \mid \mathbf{z}_{d,t,n} = k) \sim \mathrm{Multinomial}(\boldsymbol{\beta}_{k,t})$.

Note that prior studies have also explored the incorporation of covariates into dynamic topic models. Among them, two notable works are the dynamic linear topic model (DLTM, Glynn et al. 2019) and the heterogeneous classifier-free dynamic topic model (HCF-DTM, Ye et al. 2024). Although both works consider the effect of covariates, they differ from our proposed CDTM in several important ways. Specifically, DLTM and CDTM adopt different model formulations and estimation procedures. One notable distinction is that CDTM accounts for correlations among topic proportions, which are ignored in DLTM. Compared with HCF-DTM, the key distinction of CDTM lies in its assumption of dynamic topic-word distributions. Furthermore, we provide a theoretical analysis to support the validity of CDTM, as studied in Section 3.3. A more detailed comparison between these models is present in Appendices S3.1 and S3.2 of the supplementary materials.

3.2. Model Estimation

Let $\mathbf{X} = \{\boldsymbol{x}_{d,t}, 1 \le d \le D_t, 1 \le t \le T\}$, the covariate effect $\boldsymbol{\Gamma} = \{\boldsymbol{\Gamma}_t : 1 \le t \le T\}$, and the *prior mean of the covariate effect* $\tilde{\boldsymbol{\Gamma}} = \{\tilde{\boldsymbol{\gamma}}_k : 1 \le k \le K\}$. We collect all the parameters representing document-topic probabilities as $\boldsymbol{\eta} = \{\eta_{d,t,k} : 1 \le d \le D, 1 \le t \le T, 1 \le k \le K\}$ (before softmax transformation) and $\boldsymbol{\theta} = \{\theta_{d,t,k} : 1 \le d \le D, 1 \le t \le T, 1 \le k \le K\}$ (after softmax transformation). Similarly, we collect all the parameters representing topic-word probabilities as $\boldsymbol{\tau} = \{\tau_{k,t,v} : 1 \le k \le$

$K, 1 \leq t \leq T, 1 \leq v \leq V\}$ (before softmax transformation) and $\boldsymbol{\beta} = \{\beta_{k,t,v} : 1 \leq k \leq K, 1 \leq t \leq T, 1 \leq v \leq V\}$ (after softmax transformation). Further, we denote all observed words by $\mathbf{w} = \{\mathbf{w}_{d,t,n} : 1 \leq n \leq N_{d,t}, 1 \leq t \leq T, 1 \leq d \leq D\}$ and all topic indicators by $\mathbf{z} = \{\mathbf{z}_{d,t,n} : 1 \leq n \leq N_{d,t}, 1 \leq t \leq T, 1 \leq d \leq D\}$. All the variance hyper-parameters are denoted by $\boldsymbol{\Lambda} = \{\sigma_k^2, \Sigma_t, \sigma_\tau^2\}$. Given the above notations, we can derive the joint distribution of interested variables as follows,

$$
\begin{aligned}
p(\boldsymbol{\eta}, \mathbf{z}, \boldsymbol{\tau}, \boldsymbol{\Gamma}, \tilde{\boldsymbol{\Gamma}} \mid \mathbf{w}, \mathbf{X}, \boldsymbol{\Lambda}) &\propto p(\boldsymbol{\tau} \mid \boldsymbol{\Lambda}) p(\tilde{\boldsymbol{\Gamma}}) \\
&\times \left[\prod_{t=1}^{T} \prod_{d=1}^{D_t} \text{Normal}(\boldsymbol{\eta}_{d,t} \mid \boldsymbol{x}_{d,t} \boldsymbol{\Gamma}_t, \boldsymbol{\Lambda}) \right. \\
&\times \prod_{k=1}^{K} \prod_{t=1}^{T} \text{Normal}(\boldsymbol{\gamma}_{k,t} \mid \tilde{\boldsymbol{\gamma}}_k, \boldsymbol{\Lambda}) \left\{ \prod_{n=1}^{N_{d,t}} \text{Multinomial}(\mathbf{z}_{d,t,n} \mid \boldsymbol{\theta}_{d,t}) \right. \\
&\times \left. \left. \text{Multinomial}(\mathbf{w}_{d,t,n} \mid \mathbf{z}_{d,t,n}, \boldsymbol{\beta}_{\mathbf{z}_{d,t,n},t}) \right\} \right].
\end{aligned} \tag{3}
$$

Note that integrating covariates into topic models introduces inferential challenges. This is primarily due to the loss of conjugacy in the posterior distribution. That is, the normal distribution, employed to model covariate-dependent topic proportions, lacks conjugacy with the multinomial likelihood used for topic-word distributions. As a result, exact posterior inference becomes analytically intractable. Two typical approaches are commonly used to address this issue: one is the variational inference (e.g., Blei and Lafferty 2007) and the other is the MCMC method (e.g., Glynn et al. 2019). Since the MCMC method is computationally prohibitive for large-scale datasets, we choose to adopt the variational inference method implemented via an EM algorithm.

The core of the variational EM is to find a more simplified distribution that approximates (3) by breaking the relationships of certain variables. Here approximations are made for $\boldsymbol{\theta}$ and $\boldsymbol{\beta}$. We introduce three variational distributions $q(\boldsymbol{\eta})$, $q(\mathbf{z})$ and $q(\boldsymbol{\tau})$. The original distribution of $\boldsymbol{\eta}_{d,t}$ is conditioned on $\boldsymbol{x}_{d,t}$, $\boldsymbol{\Gamma}_t$, and $\boldsymbol{\Sigma}_t$. To simplify this distribution, the variational distribution $q(\boldsymbol{\eta}_{d,t}) = (\eta_{d,t,1}, \ldots, \eta_{d,t,K})^\top$ is a normal distribution with mean $\boldsymbol{\lambda}_{d,t}$ and covariance matrix $\boldsymbol{\nu}_{d,t}$, that is, $q(\boldsymbol{\eta}_{d,t}) = \text{Normal}(\boldsymbol{\lambda}_{d,t}, \boldsymbol{\nu}_{d,t})$. The original distribution of topic indicators $\mathbf{z}_{d,t} = (\mathbf{z}_{d,t,1}, \ldots, \mathbf{z}_{d,t,N_d})^\top$ is conditioned on $\boldsymbol{\eta}_{d,t}$. We assume that its variational posterior $q(\mathbf{z}_{d,t,n})$ is a multinomial distribution with parameter $\boldsymbol{\phi}_d$. Then, $\boldsymbol{\phi}_{d,t}$ serves as the variational parameter for $\boldsymbol{\theta}_{d,t}$. For topic-word probabilities, the original distribution of $\boldsymbol{\tau}_{k,t}$ is conditional on $\boldsymbol{\tau}_{k,t-1}$. To simplify this distribution, we assume that the variational posterior $q(\boldsymbol{\tau}_{k,t})$ is a normal distribution with mean $\boldsymbol{\upsilon}_{k,t}$ and covariance matrix $\boldsymbol{\varsigma}_t I_V$, that is, $q(\boldsymbol{\tau}_{k,t}) = \text{Normal}(\boldsymbol{\upsilon}_{k,t}, \boldsymbol{\varsigma}_t I_V)$. We denote $\boldsymbol{\zeta}_{k,t}$ as the variational parameter of $\boldsymbol{\beta}_{k,t}$, which is the softmax transformation of $\boldsymbol{\upsilon}_{k,t}$. Using the variational distributions, the schematic diagram of CDTM is shown in Figure 2.

Combining the variational distributions, the joint variational distribution of CDTM is $q(\boldsymbol{\eta}, \mathbf{z}, \boldsymbol{\tau}) = \prod_t \prod_d q(\boldsymbol{\eta}_{d,t}) q(\mathbf{z}_{d,t}) \prod_k q(\boldsymbol{\tau}_{k,1}, \ldots, \boldsymbol{\tau}_{k,T})$. Then, the difference between the joint variational distribution and true distribution (3) can be evaluated using the Kullback–Leibler (KL) divergence as follows,

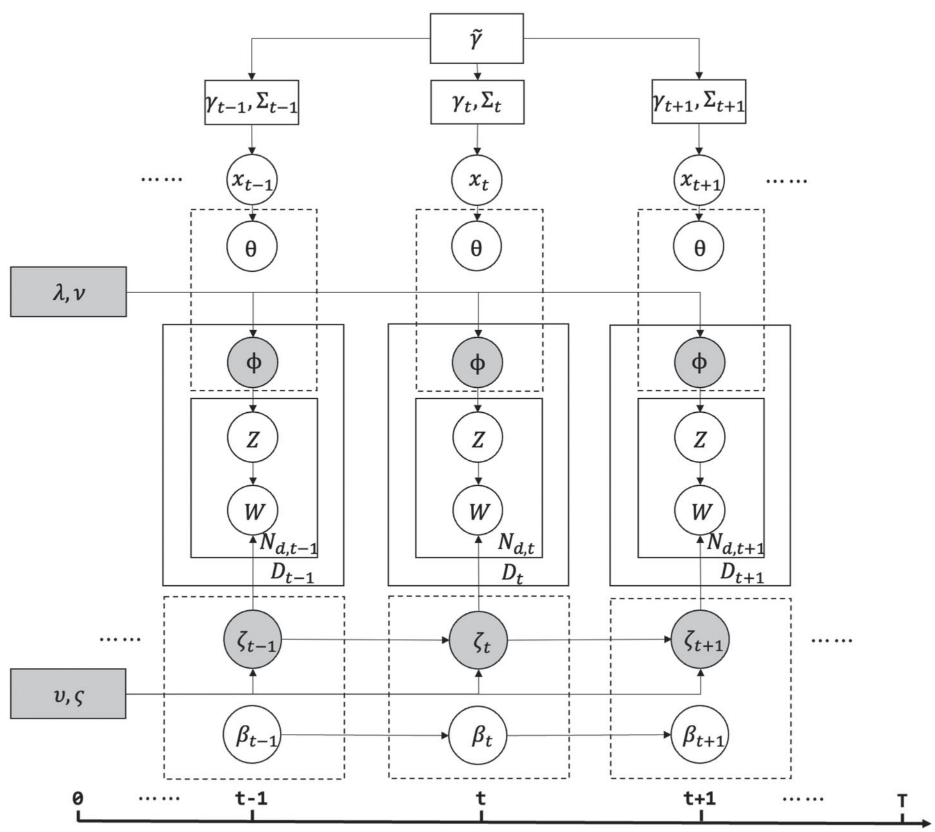


Figure 2. Schematic diagram of variational inference for CDTM.

$$
\begin{aligned}
&\mathrm{KL}\left\{q(\boldsymbol{\eta},\mathbf{z},\boldsymbol{\tau})\|p(\boldsymbol{\eta},\mathbf{z},\boldsymbol{\tau},\boldsymbol{\Gamma},\tilde{\boldsymbol{\Gamma}}\mid\mathbf{w},\mathbf{X},\boldsymbol{\Lambda})\right\}\\
=&E_q\left\{\log\prod_t\prod_d q(\boldsymbol{\eta}_{d,t})q(\mathbf{z}_{d,t})\prod_k q(\boldsymbol{\tau}_{k,1},\ldots,\boldsymbol{\tau}_{k,T})\right\}\\
&-E_q\left\{\log p(\boldsymbol{\eta},\mathbf{z},\boldsymbol{\tau},\boldsymbol{\Gamma},\tilde{\boldsymbol{\Gamma}}\mid\mathbf{w},\mathbf{X},\boldsymbol{\Lambda})\right\} \qquad (4)\\
=&E_q\left\{\log\prod_t\prod_d q(\boldsymbol{\eta}_{d,t})q(\mathbf{z}_{d,t})\prod_k q(\boldsymbol{\tau}_{k,1},\ldots,\boldsymbol{\tau}_{k,T})\right\}\\
&-E_q\left\{\log p(\boldsymbol{\eta},\mathbf{z},\boldsymbol{\tau},\boldsymbol{\Gamma},\tilde{\boldsymbol{\Gamma}},\mathbf{w},\mathbf{X},\boldsymbol{\Lambda})\right\}+\log\left\{p(\mathbf{w},\mathbf{X},\boldsymbol{\Lambda})\right\}.
\end{aligned}
$$

Based on the KL divergence derived above, we obtain the corresponding *evidence lower bound* (ELBO), which we maximize using an EM algorithm to optimize the model. The detailed derivations of the variational inference procedure via EM and the discussion of computational complexity can be found in Appendix S1 of the supplementary materials. The finite sample performance of CDTM is demonstrated via a simulation study; see Appendix S4 of the supplementary materials for details.

3.3. Theoretical Properties

In this section, we provide theoretical properties of the variational inference method for CDTM. Particularly, we aim to demonstrate that under some regular conditions, the point estimates of model parameters will converge to the true parameters as the corpus size tends to infinity. We generally follow the framework of Pati, Bhattacharya, and Yang (2018) and Yang, Pati, and Bhattacharya (2020), which study the properties of variational Bayesian under the assumption that there exist true parameters. The main idea is to establish the variational Bayesian risk upper bound under a finite-sample setting, and then derive the convergence rate.

We measure approximation quality using the squared Hellinger distance between the marginal data likelihoods $p_{\boldsymbol{\delta}}$ and $p_{\boldsymbol{\delta}^*}$: $h^2(p_{\boldsymbol{\delta}},p_{\boldsymbol{\delta}^*}) = \int(\sqrt{p_{\boldsymbol{\delta}}(y)}-\sqrt{p_{\boldsymbol{\delta}^*}(y)})^2\,dy$. Here, $\boldsymbol{\delta}$ denotes the model parameters $\boldsymbol{\delta} = (\boldsymbol{\Gamma},\boldsymbol{\tau})$, and $\boldsymbol{\delta}^*$ denotes their true values; $(\mathbf{z},\boldsymbol{\eta})$ are regarded as latent variables. Let $\widehat{q}(\boldsymbol{\delta})$ be the variational distribution that minimizes the KL divergence within our variational family. We consider the integrated Hellinger risk

$$\int h^2(p_{\boldsymbol{\delta}},p_{\boldsymbol{\delta}^*})\,\widehat{q}(\boldsymbol{\delta})\,d\boldsymbol{\delta}. \qquad (5)$$

Since $h^2(\cdot,\cdot)$ is convex in its first argument, Jensen's inequality yields

$$h^2(p_{\widehat{\boldsymbol{\delta}}_{VB}},p_{\boldsymbol{\delta}^*}) \le \int h^2(p_{\boldsymbol{\delta}},p_{\boldsymbol{\delta}^*})\,\widehat{q}(\boldsymbol{\delta})\,d\boldsymbol{\delta}, \qquad (6)$$

where $\widehat{\boldsymbol{\delta}}_{VB} = \int \boldsymbol{\delta}\,\widehat{q}(\boldsymbol{\delta})\,d\boldsymbol{\delta}$ is the variational posterior mean produced by the variational inference. Therefore, controlling the right-hand side of (6) implies consistency of $\widehat{\boldsymbol{\delta}}_{VB}$. We summarize the result below.

Theorem 1. Assume that each document has the same word count N and denote the number of documents by D. Suppose that $KpT = o(D/\log D)$ and $TKV = o(DN/\log(DN))$ as $D,N \to \infty$. Then the integrated Hellinger risk converges to 0 in probability:

$$\int h^2(p_{\boldsymbol{\delta}},p_{\boldsymbol{\delta}^*})\,\widehat{q}(\boldsymbol{\delta})\,d\boldsymbol{\delta} \xrightarrow{\text{pr}} 0. \qquad (7)$$

The detailed assumptions and proof of Theorem 1 are provided in Appendix S2. The theorem shows that the integrated Hellinger risk tends to zero under the stated rate conditions. By (6), the theorem implies that the variational posterior mean is a consistent estimator for the true global parameters.

4. Statistical Knowledge Trend Detection

4.1. Topic Modeling with CDTM

To investigate the emerging knowledge trend in statistics, we apply the proposed CDTM on the statistical publication dataset. To detect the dynamic changes in topics, the dataset is divided into 45 time slices according to the publishing year from 1980 to 2024. We consider two document-level covariates in $\mathbf{X}$ that might influence the topics represented by the dataset. The first one is the *publication venue*, which has seven levels. The second one is the *number of authors* in each paper, which is a continuous variable. Because the conference papers are only available in 2014–2024, we compute the pseudo-inverse of the covariance matrix of $\mathbf{X}$ during model fitting to prevent numerical instability and facilitate parameter updates within the variational EM algorithm. The results reveal that coefficients corresponding to conferences lacking data (in earlier periods) are consistently near zero. Other hyperparameter setups are as follows. The variance of topic distributions is set as $\sigma_\tau^2 = 0.05$. The hyperparameter σ_k^2 is independently generated from an inverse-gamma distribution Inver-Gamma(1,1). The initial value of the covariance matrix Σ_t is set as an identity matrix and updated by the variational EM iteration.

To evaluate the model performance, we consider three commonly used metrics. The first two metrics are *coherence score* (CS, Mimno et al. 2011) and *normalized pointwise mutual information* (NPMI, Aletras and Stevenson 2013), which aim to measure the semantic coherence of the top words in each topic. A higher CS or NPMI indicates that the words within a topic are more semantically related and interpretable. The third metric is *refined diversity Jaccard index* (RDJ, Deng, Siersdorfer, and Zerr 2012), which quantifies topic diversity by assessing the degree of word overlap across different topics. A lower RDJ score reflects better topic distinctiveness and less redundancy among topics. The detailed definitions of the three metrics are given below. Take topic k at time t as an example. Assume its top M words with the highest probabilities are $\boldsymbol{W} = (w_1,w_2,\ldots,w_M)^\top$. We define $F(w)$ as the number of documents including word w and $F(w,w')$ as the number of documents including both words w and w'. Coherence score and NPMI are defined as follows,

$$\mathrm{CS}(k,t;\mathcal{C}) = \frac{2}{M(M-1)}\sum_{1\le i<j\le M}\log\frac{F(w_i,w_j)+1/D_t}{F(w_j)}, \qquad (8)$$

$$\mathrm{NPMI}(k,t;\mathcal{C}) = \frac{2}{M(M-1)}\sum_{1\le i<j\le M}\log\frac{F(w_i,w_j)}{F(w_i)F(w_j)}. \qquad (9)$$

Table 4. Topic metrics under different topic numbers.

K	3	4	5	6	7	8	9	10
CS ↑	−1.879	−1.872	−1.817	−1.795	−1.765	**−1.688**	−1.732	−1.712
NPMI ↑	0.063	0.073	0.085	0.100	0.113	**0.144**	0.131	0.143
RDJ ↓	0.103	0.090	0.089	0.075	0.073	0.069	0.068	**0.066**

NOTE: The upward arrow indicates that a larger value of the metric is better, while the downward arrow indicates the opposite. The bold values indicate the best results. For metrics CS and NPMI (the larger the better), the bold values are the largest ones; while for the metric RDJ (the smaller the better), the bold values are the smallest ones.

Table 5. Topic metrics under different topic models with $K = 8$.

Model	CDTM	DTM	STM	DLTM	BERTopic	FASTopic	ProdLDA	DETM
CS ↑	**−1.688**	−1.876	−1.725	−1.963	−4.143	−1.902	−2.733	−2.140
NPMI ↑	**0.144**	0.081	0.103	0.053	0.080	−0.114	0.063	−0.010
RDJ ↓	**0.069**	0.171	0.182	0.072	0.091	0.097	0.124	0.080

NOTE: The upward arrow indicates that a larger value of the metric is better, while the downward arrow indicates the opposite. The bold values indicate the best results. For metrics CS and NPMI (the larger the better), the bold values are the largest ones; while for the metric RDJ (the smaller the better), the bold values are the smallest ones.

Let $T_{k,t}$ be the set of top M words for topic k at time t. Then RDJ is defined as

$$\mathrm{RDJ}(t; \mathcal{C}) = \frac{2}{K(K-1)} \sum_{1 \le k < k' \le K} \frac{|T_{k,t} \cap T_{k',t}|}{|T_{k,t} \cup T_{k',t}|}. \tag{10}$$

When calculating the metrics for a topic model, CS and NPMI are averaged over all topics and all time slices, and RDJ is averaged over all time slices. To choose the appropriate value of topic number, we vary K from 3 to 10. For each K, the CDTM is constructed and the metrics are calculated (Table 4). The results indicate that CS and NPMI achieve their maximum at $K = 8$, while RDJ attains its minimum at $K = 10$, with marginal improvement compared to $K = 8$. Based on this tradeoff, we finally select $K = 8$ as the optimal topic number for CDTM.

We further compare CDTM with seven competitors: the dynamic topic model (DTM, Blei and Lafferty 2006), the structural topic model (STM, Roberts, Stewart, and Airoldi 2016), the dynamic linear topic model (DLTM, Glynn et al. 2019), the BERTopic model (Grootendorst 2022), the FASTopic model (Wu et al. 2024), the ProdLDA model (Srivastava and Sutton 2017), and the dynamic embedded topic model (DETM, Dieng, Ruiz, and Blei 2019). DTM, STM, and DLTM are Bayesian topic models; DETM is a dynamic neural topic model; the BERTopic model and the FASTopic model are transformer-based neural topic models; and ProdLDA is a neural topic model based on variational autoencoders (VAEs). A detailed comparison between CDTM and these competitors can be found in Appendix S3 of the supplementary materials. Since BERTopic does not allow users to explicitly specify the number of topics, we tune the relevant parameters to yield $K = 9$, which is the most comparable setup to $K = 8$ in CDTM. All other models are applied to the real dataset with the number of topics fixed at $K = 8$. The results, summarized in Table 5, demonstrate that CDTM achieves superior performance across all three evaluation metrics.

Since CDTM can be viewed as a combination of DTM and STM to some extent, we further conduct a detailed comparison between these models. First, we summarize the key advantages and disadvantages of CDTM compared to DTM and STM. In brief, CDTM's core advantage over DTM is its explicit integration of covariates X into the generative process. Compared to STM, CDTM's primary strength lies in its ability to capture temporal dynamics. However, the advantages of CDTM come at the cost of increased model complexity, which results in longer runtime. More detailed discussions can be found in Appendix S3.8 of the supplementary materials. Second, to evaluate whether CDTM performs significantly better than DTM and STM in real data analysis, we conduct a real-data-based simulation study, in which we use LLMs to generate synthetic text that closely resembles the real data. Results show that CDTM significantly outperforms both DTM and STM in terms of the evaluation metrics; see Appendix S5.5 of the supplementary materials for details. Last, we perform a qualitative comparison of the topics generated by CDTM, DTM, and STM. For DTM, the extracted topics exhibit some similarity to those from CDTM but with lower topic quality. By using a post-hoc regression analysis, it is possible to explore the covariate effects on the extracted topics by DTM. However, the explanatory power of covariates is limited when compared to that of CDTM. As for STM, it cannot capture temporal topic dynamics. As a result, the generated topics by STM fail to reflect meaningful temporal evolution. More detailed analysis can be found in Appendices S5.6 and S5.7 of the supplementary materials.

4.2. Meanings of Extracted Topics

We focus on the meanings of extracted topics by CDTM. Notably, each topic has a probability distribution over the vocabulary of words. Following the common practice of topic models, the meaning of each topic can be summarized by analyzing the high-probability words under each topic. In addition to the estimated probability under each topic, we also consider another metric called exclusivity to summarize the topic meanings (Airoldi and Bischof 2016). Let $\widehat{\beta}_{k,t,v}$ define the estimated probability of word v under topic k at time t. The exclusivity is defined as $\text{exclusivity}_{k,t,v} = \widehat{\beta}_{k,t,v} / \sum_{j=1}^{K} \widehat{\beta}_{j,t,v}$, which reflects the relative importance of word v to the given topic k at time t.

Table 6 shows the high probability and high exclusivity words under each topic, which are summarized across the entire time span. Based on the high probability and high exclusivity words, we can summarize the meaning of each topic. We also enlist the assistance of LLMs to help summarize the meanings of topics. Specifically, we ask ChatGPT the question "Here are several key words from a series of articles in statistics. Can you help me summarize the topic of these articles in a few concise words?" and then use its response as the summary. Based on the results of Table 6, we discuss each topic as follows. More representative papers for each topic are summarized in Appendix S5.3 of the supplementary materials.

Table 6. Important words for each topic based on estimated probability or exclusivity.

	Meaning	High probability words	High exclusivity words	Summary from ChatGPT
T1	Algorithm and optimization	importance sampling, EM algorithm, convex optimization, stochastic gradient	primal dual, stochastic approximation, strongly convex, saddle, clustering algorithm	Statistical optimization
T2	Bayesian statistics	prior, posterior, uncertainty quantification, random effect, empirical Bayesian	Bayesian factor, Jeffrey prior, Laplace approximation, Gaussian process regression	Bayesian inference
T3	Deep learning	neural network, architecture, training, learning, representation, benchmark, label	GitHub, transformer, meta learning, graph neural network, pixel, transfer learning	Neural networks
T4	Biostatistics	measurement, causal effect, survival, patient, longitudinal, treatment, intervention	relative risk, breast cancer, biomarker, mediator, genetic, tumor, mortality, trial	Medical statistics
T5	Nonparametrics	density, kernel, nonparametric regression, rate of convergence, cross validation	bandwidth selection, RKHS, Euclidean space, manifold, spike, oracle inequality	Nonparametric estimation
T6	Regression	linear regression, quantile regression, asymptotic property, high dimensional, LASSO	generalized least square, sufficient dimension reduction, changepoint, Kaplan–Meier	Regression analysis
T7	Unsupervised learning	sampling, cluster, sequence, efficient, experiment, component, maximum likelihood	differential equation, gradient flow, computationally efficient, sampler, matching	Unsupervised sampling
T8	Others	matrix, block, multiple test, dependence, power, independent, robust, correlation	FDR control, correlation matrix, canonical correlation, orthogonal array, knockoff	Complex data analysis

NOTE: Based on the words, we summarize the meanings of topics. We also ask for LLMs for reference.

(T1) We summarize Topic 1 as Algorithm and Optimization, since the high-probability words include *EM algorithm*, *stochastic gradient*, and *convex optimization*. In addition, among the high-exclusivity words, some important algorithmic terms appear, such as *primal dual*, *strongly convex*, and *stochastic approximation*. One representative article of this topic is Vaswani, Yang, and Szepesvari (2022, NeurIPS).

(T2) The representative words in Topic 2 include *prior*, *posterior*, *uncertainty quantification*, *random effect*, and *Gaussian process regression*. These words are important terminologies in Bayes. Thus, we label this topic as Bayesian Statistics. One representative article of this topic is Ando (2007, BKA).

(T3) Topic 3 is mainly about Deep Learning, including some typical words such as *neural network*, *architecture*, *representation*, *transformer*, *transfer learning*, and *meta learning*. One representative article of this topic is Yang et al. (2023, NeurIPS).

(T4) Topic 4 is summarized as Biostatistics. Its representative words include *measurement*, *intervention*, *breast cancer*, *mediator*, and *mortality*. One representative article of this topic is Zhao and Zhu (2022, JASA).

(T5) Topic 5 is related to Nonparametrics. The representative words in this topic include *density*, *kernel*, *rate of convergence*, *bandwidth selection*, *manifold*, *reproducing kernel Hilbert space (RKHS)*, and *oracle inequality*. One representative article of this topic is Stéphanovitch, Aamari, and Levrard (2024, AOS).

(T6) Topic 6 is related to Regression, with its representative words including *LASSO*, *quantile regression*, *generalized least square*, and *sufficient dimension reduction*. One representative article of this topic is Cattaneo, Jansson, and Newey (2018, JASA).

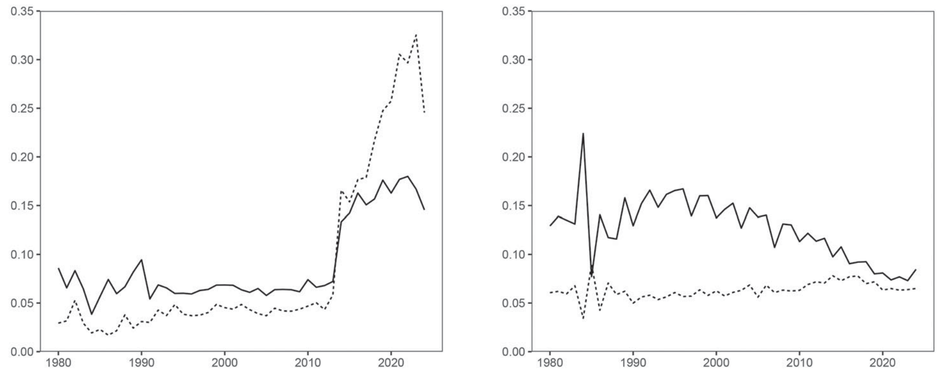
(T7) Topic 7 is summarized as Unsupervised Learning with the top words including *cluster*, *gradient flow*, *differential equation*, and *sampling*. One representative article of this topic is Kong and Chaudhuri (2021, NeurIPS).

(T8) We designate Topic 8 as Others, as it encompasses a diverse set of advanced statistical methodologies, including multiple testing, experimental design, feature screening, extreme value theory, precision matrix estimation, and others. Notably, these methodologies share a common focus on matrix-based or multivariate frameworks and are often applied to complex data structures, a relationship also noted by ChatGPT in Table 6. Representative work of this topic is Simpson, Wadsworth, and Tawn (2020, BKA).

It is worth noting that articles typically do not focus on a single topic but rather cover a mixture of topics, which is represented by $\boldsymbol{\theta}_{d,t}$ for each document. From the obtained topics, we can observe that with the advent of the AI era, statistics has increasingly focused on issues closely related to AI. For example, deep learning (T3), as a core element of modern data science, has become an emerging field of research within statistics. At the same time, many classical topics in statistics are closely connected with data science. For instance, regression analysis (T6), which is a fundamental method in statistics, is widely applied in predictive modeling within data science. Additionally, Bayesian statistics (T2) is closely related to probabilistic inference in modern machine learning methods, especially in deep learning and reinforcement learning. Therefore, we can see that classical statistical theories continue to evolve in response to the challenges posed by modern data science, driving innovation and progress across various fields. These findings are also consistent with the discussions presented in Lin et al. (2025). In the subsequent sections, we try to answer the questions (Q1)–(Q3) (see Section 2 for details) based on the results of CDTM.

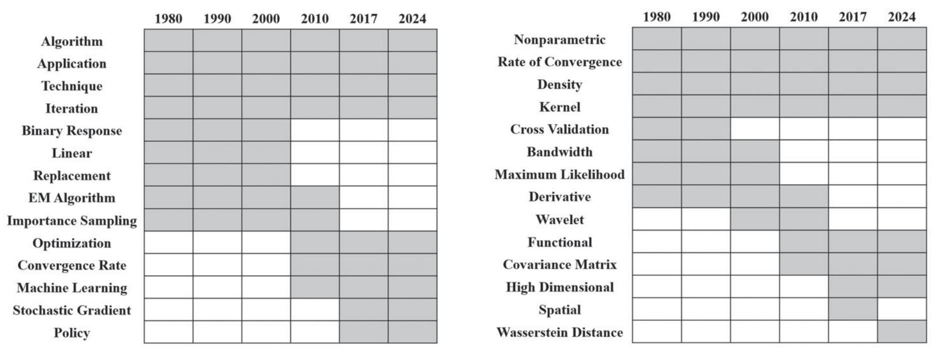
4.3. Dynamic Trend of Topics

To address (Q1) and (Q2) concerning the evolution of key topics and the field's response to the rise of AI, we analyze the temporal changes in topic proportions and word distributions. We first visualize the temporal evolution of topic proportions (i.e., $\boldsymbol{\theta}_{d,t}$) in Figure 3. The left-hand panel highlights two key trends: Algorithm and Optimization (T1), which maintains a



(a) Solid line: T1; Dotted line: T3. (b) Solid line: T2; Dotted line: T7.

Figure 3. Dynamic change of topic proportions (the *y*-axis) over time.



(a) T1 (b) T5

Figure 4. Dynamic changes in the high-probability words of Algorithm and Optimization (T1) and Nonparametrics (T5). Gray indicates that the word appears in the corresponding time period, while white indicates absence.

stable presence in statistical journals, experiences a marked surge following the inclusion of conference papers (from 2010). This pattern is even more pronounced for Deep Learning (T3). In contrast, the right-hand panel reveals diverging trajectories: Bayesian Statistics (T2) exhibits a gradual decline over time, while Unsupervised Learning (T7) steadily retains its proportional representation across the corpus.

We then discuss the dynamic changes in the high-probability words of topics. We take Algorithm and Optimization (T1) and Nonparametrics (T5) as examples, presenting their high-probability words across different time periods. Figure 4 visualizes the high-probability words for T1 and T5 across six time points: 1980, 1990, 2000, 2010, 2017, and 2024. For T1, persistent terms include *algorithm*, *technique*, and related concepts. The term *EM algorithm* dominated as a top keyword from 1980 to 2000, with foundational works such as Meng and Van Dyk (1997, JRSSB). However, the 2010s and 2020s saw a shift toward terms like *stochastic gradient*, *machine learning*, and *policy*, coinciding with advancements in deep learning and reinforcement learning. In the recent period, optimization emerged as a central theme, reflecting the increasing complexity of statistical algorithms and the need for robust optimization strategies. Key contributions include Chen, Zhang, and Wen (2022, NeurIPS) and Bian et al. (2024, JASA). In Nonparametrics (T5, Figure 4(b)), consistent terms such as *nonparametric* and *rate of convergence* underscore its focus on classical nonparametric theory. During the 1980s–1990s, dominant terms like *maximum likelihood*, *bandwidth*, and *cross-validation* aligned with traditional nonparametric methods, exemplified by Geman and Hwang (1982, AOS) and Härdle, Hall, and Marron (1992, JASA). By 2000, *wavelet* emerged as a key term, as seen in Wang (1998, JASA) and Cai and Low (2005, AOS). Recent decades reveal a pivot toward *high-dimensional* methods, with Jiang (2017, NeurIPS) and Shao, Lin, and Yao (2022, AOS) illustrating this trend. Notably, *Wasserstein distance* became prominent in 2024, which is closely related to generative models and distributional regression Chen, Lin, and Müller (2023, JASA).

Conclusion for (Q1): The evolution of statistical topics demonstrates a clear shift from classical theoretical domains toward computationally intensive and data-driven fields. As shown in Figure 3, topics like Deep Learning (T3) have surged in prominence, especially after 2010, while traditional areas like Bayesian Statistics (T2) have seen relative declines. The vocabulary evolution in Figure 4 further confirms this transition, highlighting the replacement of classical terms with concepts from machine learning and optimization.

Conclusion for (Q2): Statistics has significantly evolved in response to the rise of AI. This is evidenced not only by the high prevalence of Deep Learning (T3) and Algorithm and Optimization (T1) in recent years (as shown in Figure 3), but also by the evolution of classical topics. For instance, Nonparametrics (T5) has shifted toward the AI domain, as statistical research increasingly conducts nonparametric analysis for emerging AI concepts, such as *Wasserstein distance* in generative models (Gao

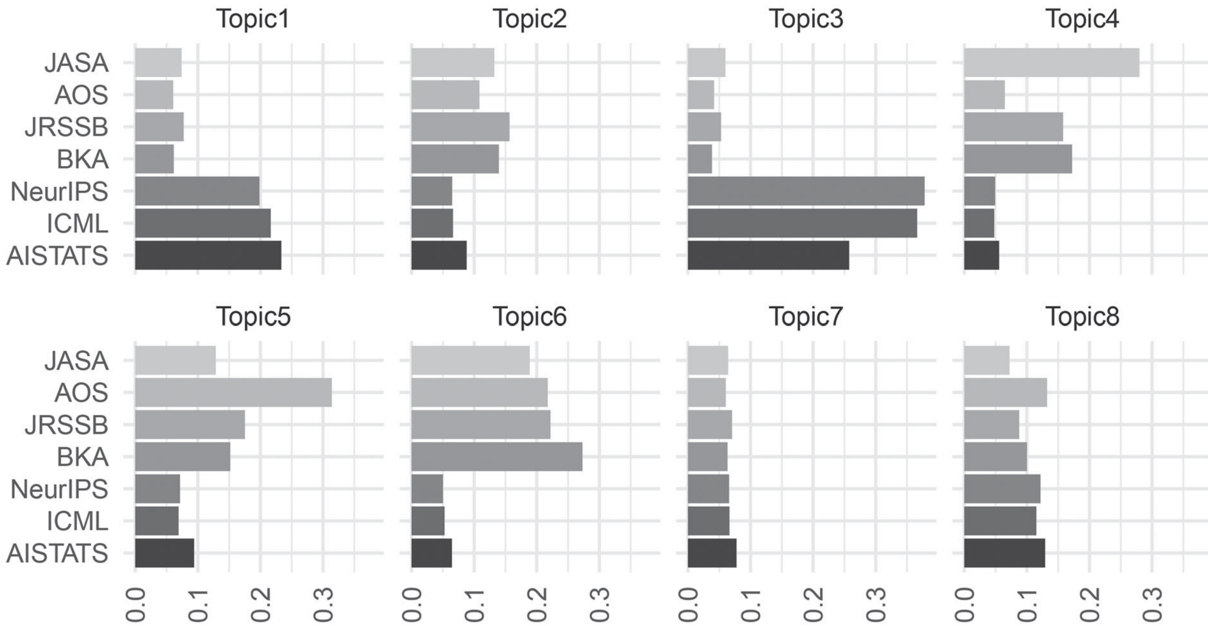


Figure 5. Proportion of topics (the x-axis) under different publication venues.

et al. 2024; Song et al. 2025). These trends highlight the deep integration of modern data science into statistical research.

4.4. Analysis of Topic Preference

To address (Q3) regarding the role of covariates in shaping statistical research trends, we first analyze topic preferences across publication venues, with the overall topic distribution displayed in Table 7. The three conferences exhibit a strong emphasis on Deep Learning (T3, 35.4%) and Algorithm and Optimization (T1, 21.0%), collectively accounting for 56.4% of their content. In contrast, the four journals prioritize Regression (T6), Nonparametrics (T5), and Biostatistics (T4), with a combined total of 59.1%. This finding reflects divergent disciplinary focuses between conference and journal publications.

We then provide a detailed analysis of topic distributions across different publication venues in Figure 5, highlighting distinct thematic preferences between publications. For example, while all three conferences prioritize Deep Learning (T3), AISTATS exhibits a smaller proportion of this topic compared to NeurIPS and ICML. Among journals, AOS diverges by emphasizing Nonparametrics (T5), which is more theoretical, over Biostatistics (T4), which is more applied; whereas JASA and BKA disproportionately concentrate on Biostatistics (T4). These patterns underscore systematic differences in thematic alignment across academic venues.

Table 7. Topic proportions over the whole corpus.

Topics	Meaning	Proportion	Journal	Conference
T1	Algorithm and optimization	12.4%	6.8%	21.0%
T2	Bayesian statistics	10.6%	13.0%	6.9%
T3	Deep learning	16.9%	4.9%	35.4%
T4	Biostatistics	12.8%	17.8%	5.0%
T5	Nonparametrics	14.6%	19.2%	7.5%
T6	Regression	15.5%	22.1%	5.3%
T7	Unsupervised learning	6.5%	6.4%	6.8%
T8	Others	10.7%	9.8%	12.1%

NOTE: The "Proportion" column refers to the proportion of the topic in the total corpus. The "Journal" and "Conference" columns refer to the proportion of the topic in the journal and conference corpus, respectively.

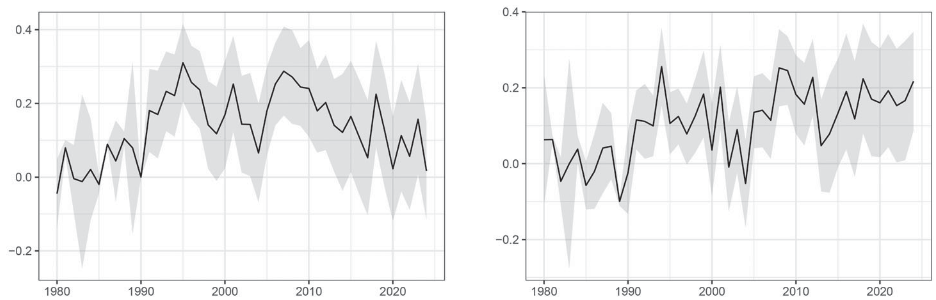
We further examine the parameter $\mathbf{\Gamma}_t$ in CDTM. In CDTM, the parameter $\mathbf{\Gamma}_t$ represents the influence of each covariate on the topic distributions at time t. We consider two covariates: the publication venue and the number of authors. To make the model identifiable, we fix $\boldsymbol{\eta}_{d,K}$ to be zero. Thus, the fitted estimates of $\mathbf{\Gamma}_t$ only include the coefficients for the first $K-1$ topics. We then perform statistical inference on $\mathbf{\Gamma}_t$. To do this, we first focus on each bunch of $\{\gamma_{t,i,j}, t = 1, \ldots, T\}$, which reflects the overall preference of covariate i over topic j across years. For each pair of (i,j), we test whether the mean of $\{\gamma_{t,i,j}, t = 1, \ldots, T\}$ is significantly nonzero or not. The corresponding results are shown in Table 8, from which we can draw several conclusions. We designate AOS as the baseline publication venue and analyze the prevalence of the remaining six venues. For Nonparametrics (T5), all other venues exhibit significantly lower preference, consistent with AOS's emphasis on theory-oriented research, such as minimax rates in nonparametric estimation. The three computer science conferences (NeurIPS, ICML, and AISTATS) demonstrate a marked preference for Algorithm and Optimization (T1) and Deep Learning (T3), aligning with the broader focus of computational research communities. JASA also shows significant favoring of these two topics, albeit with smaller coefficient magnitudes compared to the conferences. Furthermore, Deep Learning (T3) and Biostatistics (T4) correlate with larger author teams, likely reflecting the applied and labor-intensive nature of these fields, compared with more theoretical topics.

Finally, we focus on the dynamic changes of topic preferences. We take the topic preference of JASA as an example, using AOS as the baseline reference. Figure 6 illustrates the dynamic preference shifts for Algorithm and Optimization (T1) and Deep Learning (T3) in JASA. As shown, compared to AOS, JASA demonstrates a consistently positive preference for these two topics. In addition, T3 exhibits an upward trajectory, indicating

Table 8. Estimates of covariates and their significance.

		T1		T2		T3		T4		T5		T6		T7	
Publication Venues	Intercept	−0.32	***	0.02		−0.71	***	−0.64	***	1.33	***	0.97	***	0.40	***
	JASA	0.20	***	0.26	***	0.13	***	1.36	***	−1.16	***	−0.21	***	−0.55	***
	JRSSB	0.09	**	0.27	***	0.04		0.63	***	−0.91	***	−0.06		−0.48	***
	BKA	0.02		0.22	***	−0.04		0.78	***	−0.96	***	0.17	***	−0.30	***
	NeurIPS	0.96	***	−0.22	***	1.86	***	−0.48	***	−1.39	***	−1.37	***	−0.19	***
	ICML	1.04	***	−0.21	***	1.81	***	−0.49	***	−1.43	***	−1.36	***	−0.23	***
	AISTATS	1.00	***	−0.13	**	1.34	***	−0.50	***	−1.29	***	−1.32	***	−0.28	***
Number of authors		0.00		0.00		0.11	***	0.11	***	−0.05	***	−0.03	***	−0.07	***

NOTE: Baseline: T8 and AOS.
** denotes $p \leq 0.001$,
** denotes $0.001 < p \leq 0.01$ and * denotes $0.01 < p \leq 0.05$.



(a) Preference of Algorithm and Optimization (T1)

(b) Preference of Deep Learning (T3)

Figure 6. Dynamic changes of the estimated coefficient (the y-axis) for JASA with baseline as AOS. The shaded bands indicate 95% confidence intervals.

a growing emphasis of JASA on the topic of deep learning over time. More analysis for dynamic changes of preferences in different topics and publication venues can be found in Appendix S5.4 of the supplementary materials.

Conclusion for (Q3): Covariates play a critical role in shaping research trends. As detailed in Table 8, publication venues significantly dictate topic prevalence, with conferences favoring computational topics and journals adhering to theoretical foundations. Additionally, the number of authors positively correlates with applied topics like Deep Learning (T3), reflecting the collaborative nature of modern data science research.

5. Conclusion

In this work, we propose CDTM to explore the emerging knowledge trend in statistics. By analyzing the published articles in top-tier statistical journals and statistics-related conferences, we observe significant trends in the evolution of statistical research over the past decades. Specifically, we find that statistical research is increasingly aligning with the rise of AI and its growing intersection with data science. We also identify that certain covariates, such as publication venues and author collaboration patterns, significantly influence the evolution of statistical topics. By understanding these emerging trends, we can better anticipate the future directions of statistical research and foster interdisciplinary collaborations that address new challenges.

To conclude this work, we provide several interesting directions for future work. First, we currently assume the number of topics K to be fixed during different time slices. Exploring more flexible K assumptions is indeed important and will be considered in future work. Second, note that author affiliation and article keywords are valuable sources of information. Effectively incorporating them into the topic modeling process is an important direction for future research. Last, a promising future direction is to explore hybrid frameworks that combine the interpretability of CDTM with the scalability of NTMs, or to leverage LLMs to refine topic-word distributions while preserving CDTM's temporal regularization. These directions may enhance topic modeling by balancing interpretability and expressive power.

Supplementary Materials

Supplementary Materials: Proofs, simulations, detailed comparisons with other topic models, and real-data details are relegated to the Supplementary Materials (.pdf file).

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Disclosure Statement

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Data Availability Statement

The data and code are available at *https://github.com/chenxuan-he/CDTM2*.

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